

# Saturated Superfluid Vacuum VIII

## Cosmogony from the Permissive Void

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Draft

### Abstract

This paper proposes an interpretive sequence for the emergence of a saturated continuum in the Saturated Superfluid Vacuum (SSV) framework. The Permissive Void is not empty spacetime or a zero-density field configuration: it has no metric extension, proper duration, field or equation of motion. Its structurelessness cannot prohibit change, which supplies primordial temporal permissibility but does not select a first actuality. Causality and the first seed are irreducible inputs.

Actual change is proposed to establish a causal/informational screen from which metric space, physical clock rates and eventually a hydrodynamic continuum might emerge. That screen construction and the screen-to-field map are open. Only after such a continuum exists do interference, saturation and topological-defect nucleation become well-defined candidate stages.

The paper is explicitly interpretive and ontological in scope. It does not claim to present a complete quantitative cosmology. Its purpose is narrower: to state the founding logic of the cosmogonic picture clearly and separately from the quantum-mechanics derivation (Paper VII-a), the emergent-geometry derivation (Paper VII-b), and the particle-mass and force programmes (Papers I–IV). The Void is treated not as a productive mechanism but as the absence of structure and therefore as a permissive condition that cannot forbid change. Every claim in this paper is tagged *derived*, *structural*, *candidate*, or *interpretive* in the Discussion (§5), with explicit support criteria stating what observational or theoretical work would change a claim’s status. The Kibble–Zurek calculation is a continuum TDGL proxy rather than a simulation of the Void, and the dark-energy-as-background-relaxation reading is structural; neither is derived.

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# 1 Introduction

This paper presents the staged cosmogony of the Saturated Superfluid Vacuum framework in its own right. The goal is not to derive every later structure of the series again, but to separate the ontological and cosmological sequence from the quantum and gravity derivations: permissive Void, primordial temporal permissibility, first actualisation, causal screen, continuum formation, saturation, and possible defect nucleation.

The body below carries the cosmogonic content of the legacy unified SSV V draft, separated into this scoped paper. The quantum-mechanics derivation lives in Paper VII-a, the emergent-geometry derivation in Paper VII-b [1], and the particle and force content remains in Papers I–IV. Each claim in this paper is tagged in Section 5.

## 2 Cosmogony: From the Permissive Void to the Resonant Cosmos

We now ask the deepest question the framework permits: *how could a saturated superfluid become physically meaningful at all?* The following six stages separate what is defined, what is postulated and what remains open. They are an ontological sequence, not six dynamical consequences already derived from one equation.

### 2.1 Stage 1 — The Permissive Void

The *Permissive Void* is the absence of actual structure. It is not empty physical space, a thermodynamic vacuum or a field configuration. It has no metric, boundary, finite or infinite volume, proper duration, density, temperature, entropy, curvature, Hilbert space or equation of motion. In particular,  $\rho = 0$  and  $\Psi = 0$  are already structured mathematical statements and therefore belong to a later continuum description.

Because the Void has no law, barrier or state-maintaining mechanism capable of enforcing permanent stasis, it cannot prohibit change. This entails the *possibility* of change, not its occurrence:

$$\neg \text{Prevent}_{\text{Void}}(\Delta) \implies \Diamond \Delta, \quad \Diamond \Delta \not\Rightarrow \Delta.$$

The first implication is called *primordial temporal permissibility*. It is not a sequence of empty instants, elapsed duration or time measured by clocks. Saying that it is unbounded means that no temporal limit is defined, not that infinitely many seconds pass before physical time exists.

### 2.2 Stage 2 — First Actualisation and Causal Order

Permissibility does not choose an actuality. This paper therefore takes both causality and the occurrence of a first seed as irreducible inputs. It does not explain why the seed occurs, why it has one form rather than another, or whether probability language applies before a physical state space exists.

Causality gives actual change a directed influence relation. This is ordinal or causal time: before–after without a clock, duration or metric. Reality is not treated as a succession of separately existing frames or states. It is one continuously changing process; an instantaneous state or complete history is a mathematical description of that process, not an ontological object that replaces another. The past is available only through relations and records present in the continuing process, while the future denotes causally admissible possibility.

The seed is consequently called the *first actualisation* or *primordial perturbation*. “Primordial photon” may be used only as an interpretive metaphor after a condensate and its propagating modes exist. No photon, classical field perturbation or physical energy is defined at the pre-continuum stage.

### 2.3 Stage 3 — The Causal Screen and Physical Spacetime

An actual change distinguishes realised causal information from unrealised possibility. This paper calls that distinction the candidate *primordial screen*. The screen is not initially a two-dimensional surface embedded in a pre-existing three-dimensional space. Its neighbourhood structure, dimension, area and geometry remain to be derived.

Finite change is provisionally associated with update capacity. A conserved measure of that capacity is what a later clock-bearing description may call energy; neither its conservation law nor its absolute scale is derived here. Repeated causal relations on the continuously changing screen are proposed to define metric separation and local clock rates. Physical space and proper time would then be relational measures of the screen’s continuous change, not pre-existing containers. Causal orientation is fundamental; the experienced flow of time is not a physical current, and a thermodynamic arrow requires records, boundary conditions and coarse-graining.

The map from this causal screen to a continuum complex field is open. Only after such a map exists is it meaningful to introduce a domain, coordinates,  $\Psi$ ,  $\rho = |\Psi|^2$ , or the LogSE/GPE equation of Paper I [2]. Within that later description the deterministic solution  $\Psi = 0$  remains fixed unless a seed, instability or noise law is specified. Positive pressure at finite density does not itself establish such an instability. Issue #222 tracks this required distinction and construction.

### 2.4 Stage 4 — Interference and the Seed Pattern

A seed disturbance could form a standing pattern only if a reflection, scattering or boundary mechanism generates a counter-propagating component; that mechanism is not derived here. Two counter-propagating modes of wavenumber  $k$  produce a standing wave in the candidate continuum:

$$\delta\rho(\mathbf{x}, t) = \delta\rho_0 \cos(kx) \cos(\omega t), \quad (1)$$

with  $\omega = ck$  in the dilute limit. Here  $x$  and  $t$  are emergent continuum coordinates, not properties assigned to the primordial Void. This standing wave is the *seed pattern*: a spatially modulated template of alternating high- and low-density regions. Crucially, this pattern carries *geometric information* — a preferred length scale  $\lambda = 2\pi/k$  — from which all subsequent structure will grow.

### 2.5 Stage 5 — Nonlinear Growth to Saturation

The following growth sequence is a candidate scenario, not a result of the current LogSE sign choice. A dynamical calculation would have to show that positive seed regions grow while depleted regions do not and determine the corresponding nonlinear transport.

The growth proceeds through three sub-stages:

1. **Exponential amplification** (linear instability regime): the high-density crests are hypothesised to grow as  $\rho(t) \propto e^{\gamma t}$ . The estimate  $\gamma \sim c/\xi_0$  is dimensional; no instability eigenvalue has been computed.

2. **Nonlinear broadening:** as  $\rho$  approaches  $\rho_0$ , the equation of state stiffens and the growth slows. The density profile develops a plateau at  $\rho_0$  separated by sharp domain walls. Whether such surfaces support any force-sector interpretation is a separate open problem.
3. **Saturation:** when  $\rho \rightarrow \rho_0$  throughout the high-density crests, the medium has reached its ground state locally. The process is exothermic: the binding energy released goes into kinetic energy of the flow field, driving turbulence.

This stage is an analogy to inflation and reheating, not a replacement cosmology: no expansion history, e-fold count or perturbation spectrum has been derived from it.

## 2.6 Stage 6 — Vortex Nucleation: Candidate Defects

When neighbouring saturation fronts collide, phase mismatches can relax smoothly or can trap vortex filaments, depending on the field dynamics and the phase winding around a closed loop. Kibble–Zurek theory predicts a statistical defect density, not inevitable nucleation at every intersection.

SSV interprets any surviving defects as candidate precursors of its particle sector; the present calculation does not show that they relax specifically into electron rings or proton breathers. Their number density is hypothesised to follow Kibble–Zurek scaling:

$$n_{\text{defect}} \xi^3 \sim \left( \frac{\tau_0}{\tau_Q} \right)^{3\nu/(1+\nu z)}, \quad (2)$$

where  $\tau_0$  and  $\tau_Q$  are the microscopic and quench times,  $\nu$  is the correlation-length exponent, and  $z$  is the dynamic exponent.

Connecting this defect density to the baryon-to-photon ratio  $\eta \sim 6 \times 10^{-10}$  additionally requires a calculation of both baryon assignment and phonon production. Neither that mapping nor the cosmological quench time is presently derived.

## 3 Synthesis and interpretive context

### 3.1 Synthesis (interpretive)

The six stages of Section 2, considered alongside the current results and negative findings of the series, organise into the following status hierarchy. This table is a synthesis aid, not a derivation.

| Level     | Physical Object                             | SSV Identification                                                                  | Status                      |
|-----------|---------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------|
| Void      | absence of actual structure                 | temporal permissibility because change can not be prohibited                        | interpretive foundation     |
| Causality | directed influence within continuous change | primitive relation; first seed is a separate boundary fact                          | unexplained input           |
| Spacetime | causal screen                               | metric separation and proper time proposed to emerge from screen relations          | open construction           |
| Continuum | saturated medium at $\rho_0$                | corrected LogSE is a candidate constitutive description after a screen-to-field map | model mathematics; map open |
| Matter    | persistent defects                          | integer winding survives, but particle identities and masses do not follow          | candidate/open              |
| Forces    | collective interactions and geometry        | several hydrodynamic forms and negative results survive; no four-force derivation   | programme closed            |
| Arrow     | records and entropy                         | coarse-grained asymmetry within reversible microscopic dynamics                     | emergent/open               |

The table separates a permissive ontological beginning from later physical claims. In particular, corrected continuum mathematics is not evidence that the Void produces a LogSE field, and topological labels do not establish particle identities or observed masses.

### 3.2 Cosmogony as candidate self-consistency picture (interpretive)

The cosmogonic sequence does not derive a fundamental length. If a screen-to-continuum map and a counter-propagating mechanism exist, the continuum seed wavelength  $\lambda_{\text{seed}} = 2\pi/k$  would provide one candidate pattern scale. The dimensionless ratios  $\alpha$  and  $G_{\text{eff}}/(c^2\xi)$  are not derived by this paper and must not be used to turn that conditional statement into a self-tuning result. The dependencies are

1. primitive causality and an unexplained first actualisation;
2. a causal-screen construction and screen-to-field map;
3. a continuum instability, seed and counter-propagating mechanism;
4. nonlinear field dynamics that actually reaches saturation and supports persistent defects.

None of these bridges is supplied by the Void's inability to prohibit change. The sequence is therefore a candidate self-consistency picture, not a derivation of  $\rho_0$ ,  $\xi$ ,  $\alpha$ ,  $G$ , or fine-tuning.

### 3.3 Cosmological constant and dark energy

The cosmological-constant identification belongs structurally to Paper VII-b, which develops an emergent-metric candidate and the possible role of the saturation pressure  $P_0$  as a source of  $\Lambda$  ([1], §5.2). In cosmogonic language, the residual cosmological acceleration corresponds to the slow relaxation of the saturated background  $\rho_0$  at cosmological timescales rather than to an independent dark-energy sector; this is the qualitative statement. The quantitative value of  $\Lambda$  and any quantitative connection to the present-day Hubble rate  $H_0$  are open and not carried by

this paper. The follow-up closure note for issue #51 shows why the numerical route is blocked by the missing absolute saturation pressure  $P_0$ , and identifies the observational discriminator

$$\beta \equiv \frac{d \ln \Lambda_{\text{eff}}}{d \ln a}$$

for a slowly relaxing background,  $\Lambda_{\text{eff}}(a) = \Lambda_0 a^\beta$ ;  $\beta = 0$  is the  $\Lambda$ CDM limit. Neither VII-b nor this paper has established  $\Lambda$  as resolved.

## 4 Focused Candidate Programmes

Two quantitative programmes are associated with the cosmogonic picture. C1 has an exploratory continuum computation whose result is inconclusive; C2 remains uncomputed. Neither is a derived consequence of the Void or screen.

**C1. Defect-density scaling in a continuum saturation transition.** A numerical Kibble–Zurek proxy for a continuum symmetry-breaking transition produces a defect density that scales as  $n_{\text{def}} \xi^d \sim (\tau_0/\tau_Q)^{d\nu/(1+\nu z)}$  with the quench rate  $1/\tau_Q$ . The committed instrument is a stochastic TDGL model; it is not a simulation of the structureless Void, the causal screen, or a derived screen-to-LogSE transition. Status: *candidate with exploratory numerical evidence*; the finite scan performed in [kibble\\_zurek\\_gpe](#) is summarised below.

### Exploratory Kibble–Zurek test (Programme C1)

A stochastic time-dependent Ginzburg–Landau evolution on a 2D periodic grid ( $N = 160$ , box  $L = 160\xi$ ,  $\Delta t = 0.1$ , 6 seeds per quench rate, quench through  $\mu_{\text{crit}} = 0$  with  $\mu_{\text{initial}} = -1, \mu_{\text{final}} = +1$ ) and phase-winding defect count after a settling window in the saturated regime gives the defect-density scan

| $\tau_Q$ (medium units) | $\bar{n}_{\text{def}}$ (count $\pm$ std) | $n_{\text{def}}/\xi^{-2}$ |
|-------------------------|------------------------------------------|---------------------------|
| 20                      | $13.2 \pm 2.0$                           | $5.1 \times 10^{-4}$      |
| 40                      | $9.5 \pm 2.8$                            | $3.7 \times 10^{-4}$      |
| 80                      | $10.5 \pm 2.6$                           | $4.1 \times 10^{-4}$      |
| 160                     | $7.3 \pm 2.4$                            | $2.9 \times 10^{-4}$      |
| 320                     | $6.7 \pm 1.8$                            | $2.6 \times 10^{-4}$      |

An unweighted log–log fit gives  $\alpha_{2D}^{\text{fit}} = 0.234 \pm 0.052$  (regression standard error), compared with the mean-field expectation  $\alpha_{2D}^{\text{MF}} = d\nu/(1+\nu z) = 1/2$  for  $(d, \nu, z) = (2, 1/2, 2)$  and the 3D-XY analogue  $\alpha_{3D}^{\text{XY}} = 3 \times 0.6717/(1 + 1.3434) = 0.86$ . The fitted two-dimensional exponent lies about 5.1 regression standard errors below the mean-field value. The limited dynamic range and six seeds do not justify calling the expected exponent confirmed.

Cosmological identification. In SSV the natural microscopic time is  $\tau_0 \equiv \xi/c$  and the quench-rate time is the inverse Hubble parameter at the saturation epoch. The dimensionless ratio  $\tau_Q/\tau_0$  is the single SSV-unknown that fixes  $\eta$  via the KZ formula only after a defect-to-photon mapping is supplied. Inverting the observed  $\eta$  for the 3D mean-field exponent  $\alpha_{3D}^{\text{MF}} = 3/4$ , the observed  $\eta = 6 \times 10^{-10}$  requires  $\tau_Q/\tau_0 \approx 2 \times 10^{12}$ , and for  $\alpha_{3D}^{\text{XY}} = 0.86$  requires  $\tau_Q/\tau_0 \approx 5 \times 10^{10}$ . These are required input values inferred from the observation, not SSV predictions. The saturation-epoch Hubble rate and therefore  $\tau_Q/\tau_0$  have not been calculated.

Falsification status. The mean defect density trends downward with  $\tau_Q$ , but the scan does not recover the expected exponent. Programme C1 therefore remains *candidate with exploratory numerical evidence*. A meaningful test requires (i) extending the dynamic range in  $\tau_Q$  to  $\geq 10^4$  for a tight exponent fit, and (ii) fixing the cosmological identification of  $\tau_Q/\tau_0$  from a Paper VII-b emergent-metric input.

**C2. Late-time acceleration as background relaxation.** Late-time cosmological acceleration should track the slow relaxation of the saturated background  $\dot{\rho}_0/\rho_0$  rather than require an independent dark-energy sector. Falsification condition: a measured deviation from  $\Lambda$ CDM in the one-slope relaxation observable  $\beta = d \ln \Lambda_{\text{eff}} / d \ln a$ , tested coherently across  $H(z)$ , BAO distance ratios, and supernova luminosity distances. Status: *structural with identified observational discriminator*; promotion to a numerical prediction requires the VII-b / Paper IV derivation of the absolute saturation pressure  $P_0(\rho_0)$ .

## 5 Discussion: scope, status, and what would count as support

### 5.1 Claim taxonomy used in this paper

Following [numerical-minimisation-roadmap](#) in the repository, each claim in this paper is one of:

- *derived* – analytically reduced from the SSV Lagrangian and/or a converged numerical computation (Papers I, II) and *reused* here without further derivation;
- *structural* – the dependence on the framework is clear and the result is structurally consistent, but the number depends on a computation deferred to another paper or workstream;
- *candidate* – a proposed result stated here that has a definite falsification condition but the supporting computation has not yet been performed in the repository;
- *interpretive* – a synthesis or ontological statement that organises the SSV results into a single picture without claiming new quantitative content.

### 5.2 Claim status of this paper’s content

- Stages 1–3 of Section 2 (Permissive Void, first actualisation and causal screen): *interpretive*. The Void’s non-prohibition of change explains primordial temporal permissibility; causality and the first seed are primitive inputs, while screen geometry, proper time and the continuum map remain open.
- Stages 4–6 (continuum interference, growth and defect nucleation): *candidate*. The Kibble–Zurek defect-density proxy has exploratory numerical evidence after the TDGL scan reported in Programme C1 above, but the instrument starts from an already defined continuum and does not simulate the Void or causal screen.
- The synthesis table in Section 3 carries an explicit per-row status column; each row’s quantitative content comes from another paper in the series.
- The self-consistency reading (Section 3.2): *interpretive*, with explicit non-claims about the screen-to-field map, fundamental scale,  $\alpha$ ,  $G$ , and fine-tuning.

- The cosmological-constant statement (Section 3.3): *structural*, deferring the derivation to Paper VII-b and the numerical value of  $\Lambda$  to future work.
- Programmes C1 and C2 (Focused Candidate Programmes section above): C1 is *candidate with exploratory numerical evidence* (the TDGL scan has the expected trend but not the expected exponent, and the cosmological identification of  $\tau_Q/\tau_0$  is uncomputed); C2 is *structural* with explicit falsification conditions.

### 5.3 What observational or theoretical support would matter

1. *A continuum saturation-transition proxy.* Performed in [kibble\\_zurek\\_gpe](#): TDGL evolution on a  $160 \times 160$  periodic grid produces phase-winding defects in the saturated regime, with the count scaling monotonically with the inverse quench rate at fitted exponent  $\alpha_{2D}^{\text{fit}} = 0.234 \pm 0.052$ , about 5.1 regression standard errors below the mean-field KZ prediction  $\alpha_{2D}^{\text{MF}} = 0.5$ . The cosmological extrapolation to  $\eta$  requires at least the dimensionless input  $\tau_Q/\tau_0$ ; values  $\tau_Q/\tau_0 \in [5 \times 10^{10}, 2 \times 10^{12}]$  are obtained by inverting the observed  $\eta = 6 \times 10^{-10}$  for the 3D-XY and mean-field exponents respectively; they are not independently predicted. Promotion of programme C1 requires (i) extending the dynamic range in  $\tau_Q$  to  $\geq 10^4$  for a tight exponent fit or (ii) fixing the cosmological  $\tau_Q/\tau_0$  identification from a Paper VII-b emergent-metric input. Even a successful continuum result would not establish the prior causal screen or its map to the field.
2. *The issue #51 relaxing- $\rho_0$  observable programme.* The numerical value of  $\Lambda$  remains underdetermined until VII-b / Paper IV derive the absolute saturation pressure  $P_0(\rho_0)$ , but C2 now has a concrete empirical discriminator: fit  $\beta = d \ln \Lambda_{\text{eff}} / d \ln a$  in  $\Lambda_{\text{eff}}(a) = \Lambda_0 a^\beta$  and require the same slope to account for the correlated residuals in  $H(z)$ , BAO  $D_M(z)$ , and supernova  $D_L(z)$ . A future derivation of  $P_0(\rho_0)$  would turn this discriminator into a numerical SSV prediction.
3. *A screen-to-continuum construction.* This requires one causal state space, a continuous law of change, a conserved quantity that becomes energy, relational dimension and metric, physical clock rates, and a derivation of the continuum order parameter without inserting the desired LogSE dynamics.
4. *A no-go or partial-go result for the first physical continuum seed.* The structureless Void has no thermodynamic equilibrium or field perturbations. The relevant question begins only after a screen-to-continuum map exists: can that continuum support an instability or seed law without inserting the desired perturbation? The present paper leaves this open.

### 5.4 What this paper is not

This paper does not derive: a complete quantitative cosmology; a CMB spectrum; a primordial gravitational-wave background; a baryogenesis mechanism or defect-to-baryon mapping; the numerical value of  $\Lambda$ ; or the fine-structure constant  $\alpha$ . Each of those belongs to another paper or future workstream. Treating this paper as the final manifesto for the SSV programme would re-introduce the catch-all problem that the split into separate papers VII-a, VII-b, and VIII was designed to avoid.

## 6 Conclusion

Within the scoped SSV programme, this paper supplies an interpretive cosmogony rather than a derived creation dynamics. The structureless Void cannot prohibit change and therefore admits primordial temporal permissibility, while causality and the first actualisation remain irreducible inputs. A causal/informational screen is proposed as the bridge to physical space and proper time, but its geometry and continuum map are open. The subsequent saturation and defect sequence begins only after that missing bridge and remains candidate-grade.

## References

- [1] S. Norland, *Saturated Superfluid Vacuum VII-b: Emergent Geometry and the Dissolution of Quantum Gravity*, SSV programme, forthcoming.
- [2] S. Norland, *Saturated Superfluid Vacuum I: Topological Defects in a Saturated Superfluid Vacuum — The Geometric Origin of Matter*, 10.5281/zenodo.19651986, 2026.

## A Code and Issue References

Each reference below is pinned so it can be checked directly against the repository (<https://github.com/StigNorland/SVT>): issues link to their tracker entry, and each script or report links to the exact version — the commit that last modified it. Issue numbers in the text hyperlink to this list.

### Issues.

- #51 — <https://github.com/StigNorland/SVT/issues/51>
- #222 — <https://github.com/StigNorland/SVT/issues/222>

### Code (each pinned to the commit that last modified it).

- `instruments/paper_viii/kibble_zurek_gpe.py` @f204600 — [https://github.com/StigNorland/SVT/blob/f204600/instruments/paper\\_viii/kibble\\_zurek\\_gpe.py](https://github.com/StigNorland/SVT/blob/f204600/instruments/paper_viii/kibble_zurek_gpe.py)

### Reports (result notes, pinned to their last commit).

- `docs/numerical-minimisation-roadmap.md` @77eff54 — <https://github.com/StigNorland/SVT/blob/77eff54/docs/numerical-minimisation-roadmap.md>

**Change record.** This paper states its *current* status only. Corrections, withdrawals and re-framings are recorded in <https://github.com/StigNorland/SVT/blob/main/papers/SSV-VIII/CHANGELOG.md>, and the full history is in the repository’s commits.